

*MA.912.AR.3.1***Benchmark**

MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.

Benchmark Clarifications:

Clarification 1: Within the Algebra I course, instruction includes the concept of non-real answers, without determining non-real solutions.

Clarification 2: Within this benchmark, the expectation is to solve by factoring techniques, taking square roots, the quadratic formula and completing the square.

Connecting Benchmarks/Horizontal Alignment

- MA.912.AR.1.2, MA.912.AR.1.3, MA.912.AR.1.7

Terms from the K-12 Glossary

- Coefficient
- Quadratic Equation
- Real Numbers
- x -intercept

Vertical Alignment**Previous Benchmarks**

- MA.8.AR.2.3

Next Benchmarks

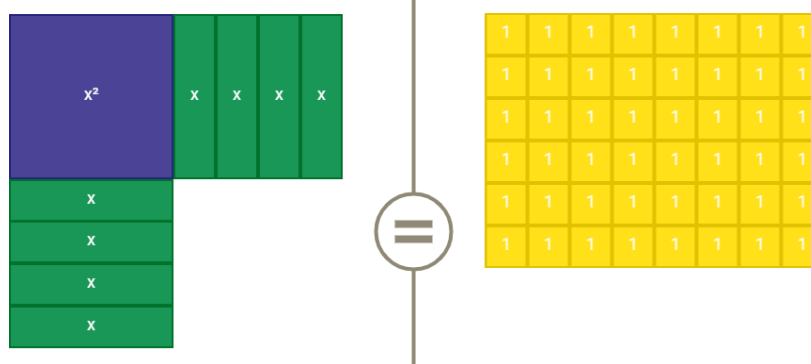
- MA.912.AR.3.2

Purpose and Instructional Strategies

In grade 8, students solved quadratic equations in the form of $x^2 = p$. In Algebra I, students solve quadratic equations in one variable over the real number system. In later courses, students will solve quadratic equations in one variable over the real and complex number systems.

- Instruction includes the use of manipulatives, models, drawings and various methods, including Loh's method.
- Instruction allows the flexibility to solve quadratics using factoring techniques, taking the square root, using the quadratic formula and completing the square. Students should understand that one method may be more efficient than another depending on the content and context of the problem (MTR.2.1).
- Instruction emphasizes the understanding that solving a quadratic equation in one variable is the same as the process of determining x -intercepts, or roots, of the graph of a quadratic function.
- While the derivation of the quadratic formula is not an expectation of this benchmark, students can develop the quadratic formula by using completing the square to isolate x in the equation $ax^2 + bx + c = 0$; making the connection to solving literal equations.

- Instruction includes evaluating the discriminant ($b^2 - 4ac$) to determine whether there is one real solution (equals zero), two real solutions (equals a positive rational number) or two complex solutions (equals a negative rational number).
 - Discuss the connection of the number of solutions of a quadratic equation to the graph of a quadratic function. Guide students to see that real solutions result in roots or x -intercepts and that quadratic functions that do not produce real roots never touch the x -axis (MTR.5.1).
- Instruction on completing the square includes the use of algebra tiles or area model drawings. Students should understand the visual nature of completing the square and connect it to their prior work with area models.
 - For example, consider $x^2 + 8x = 48$ and solve by factoring to show that the two solutions are $x = -12$ and $x = 4$. Then, ask the question “If we look at the binomial $x^2 + 8x$ on the left, what would we have to add to it to make a perfect square trinomial?” Write the equation $x^2 + 8x + \quad = 48 + \quad$ on the board to represent the question. Now, represent the equation using algebra tiles or by drawing an area model as shown below.



- Lead student discussion about the quantity of 1 unit by 1 unit tiles needed to “complete the square” on the left hand side. Once they state the need for 16 tiles, point out that 16 tiles must also be added to the right to maintain equivalence then write the number 16 into both blank boxes. Students can then factor the perfect square trinomial to arrive at $(x + 4)^2 = 64$. Have student solve using square roots to find the same solutions of $x = -12$ and $x = 4$ (MTR.2.1, MTR.4.1, MTR.5.1).
- In many contexts, students may generate solutions that may not make sense when placed in context. Be sure students assess the reasonableness of their solutions in terms of context to check for this (MTR.6.1).
 - For example, the time it takes for a ball to drop from a height of 28 feet can be modeled by $0 = -16t^2 + 28$. Students solve this equation to find that $t \approx \pm 1.32$ seconds. Through discussion, students should see that -1.32 seconds does not make sense in context and therefore should be omitted.

- Enrichment of this benchmark includes determining if a quadratic is a perfect square trinomial.
 - For example, given the equation $0 = 4x^2 - 12x + 9$, students can identify a , b , and c as 4, -12 and 9, respectively. Students should recognize that a and c are perfect squares; where a is $2^2 = 4$ and c is $(-3)^2 = 9$. Since the coefficient of b is twice the product of the square roots ($-12 = 2(2)(-3)$), it can be determined that the given equation is a perfect square trinomial.

Common Misconceptions or Errors

- When completing the square, many students forget to use the addition or subtraction property of equality to add or subtract values from the other side of the equation. Remind these students that additions or subtractions from one side of an equation must be replicated on the other to maintain equivalency.
- When completing the square and removing a common factor from the x^2 and x term, students may forget to consider that factor when adding/subtracting from the other side of the equation.
 - For example, when solving $10 = 12t^2 + 48t + 55$, students may ultimately add 4 rather than 48 to the left side. Help students to see that there are 12 sets of 4 ultimately being added to the right and therefore, there must be twelve sets added to the left as well.
- Some students may see equations such as $x^2 - 4x + 6 = 27$ and use the number 2, -4 and 6 as a , b and c , respectively, in the quadratic formula, arriving at incorrect solutions. In these cases, graph the related function and ask students if the solutions they calculated correspond to the roots of the parabola. Once they see they do not, have students set the equation equal to zero and recalculate.

Strategies to Support Tiered Instruction

- When solving using the quadratic formula, instruction includes separating the two possible solutions to two equations.
 - For example, when determining the values of x in the equation $6x^2 - 17x + 12 = 0$, students can set up the two equations using the quadratic formula: $x = \frac{-(-17) \pm \sqrt{(-17)^2 - 4(6)(12)}}{2(6)}$.
- Teacher provides guided notes that show step-by-step directions for solving problems.
 - For example, teacher can use the directions below when using the quadratic formula.

Determine the value(s) of x in the equation.	$6x^2 - 17x = 12$
Identify the a , b and c values in the equation.	$6x^2 - 17x - 12 = 0$ $a = 6, b = -17, c = -12$
Substitute a , b and c into the quadratic formula.	$x = \frac{-(-17) \pm \sqrt{(-17)^2 - 4(6)(-12)}}{2(6)}$

- Teachers can remind students when completing the square that additions or subtractions from one side of an equation must be replicated on the other side to maintain equivalency.

- When addressing removing a common factor for completing the square, instruction includes highlighting the leading coefficient.
 - For example, when solving $10 = 12t^2 + 72t + 55$, students may add 9 rather than 108 to the left side in the completing square step. Help students to see the leading coefficient of 12 indicates there are 12 sets of 9 being added to the right and therefore, there must be twelve sets added to the left as well.
- Instruction should include equations with terms on both sides of the equation. Remind students to put the equation in standard form before using the quadratic formula.
- Teacher co-creates a graphic organizer, such as the one below, to include different methods used to solve quadratics given specific forms.

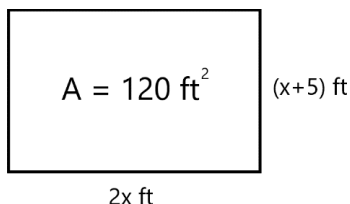
Given Quadratic	Potential Method to Solve
Standard form where a , b , c are large and or non-integer rational numbers	Quadratic Formula Completing the Square Loh's Method
Standard form where one set of factors for the product of a and c equal b Two term quadratic with the x^2 and x terms only	Factoring Completing the Square
Perfect Square Trinomial	Factoring Perfect Square Trinomial
Two term quadratic with the x^2 term and constant only Vertex form	Taking Square Root

Instructional Tasks*Instructional Task 1 (MTR.2.1)*

Given the equation $x^2 + 6x = 13$, what value(s) of x satisfy the equation?

Instructional Task 2 (MTR.3.1)

Given the figure below, write an equation that could be used to determine the length and width of the rectangle.

*Instructional Task 3 (MTR.3.1)*

Given the quadratic equation $x^2 + 6x = 13$.

- Part A. Discuss the method or strategy you would use to find the solution(s). Explain why you chose that method.
- Part B. How do you determine the number of solutions for the given equation? How do you know if the solutions are real or non-real?
- Part C. Find the x-intercepts of the quadratic equation. What do the x-intercepts tell you about the equation's graph?
- Part D. Work with a partner to create a real-world context for the equation.
- Part E. How could you rewrite the given equation so that there is only one solution?

Instructional Items*Instructional Item 1*

Devonte throws a rock straight down off the edge of a cliff that overlooks the ocean. The distance (d) the rock falls after t seconds can be represented by the equation $d = 16t^2 + 24t$. If the ocean's surface is 16.4 feet below the cliff, to the nearest tenth, how many seconds will it take for the rock to hit the ocean's surface?

Instructional Item 2

What are the solutions to the equation $-0.25x^2 + 4x = 0.75$. Round to the nearest tenth if necessary.

Instructional Item 3

What are the exact solutions to the equation $5x^2 - \frac{17}{2}x + \frac{3}{2}$?

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*